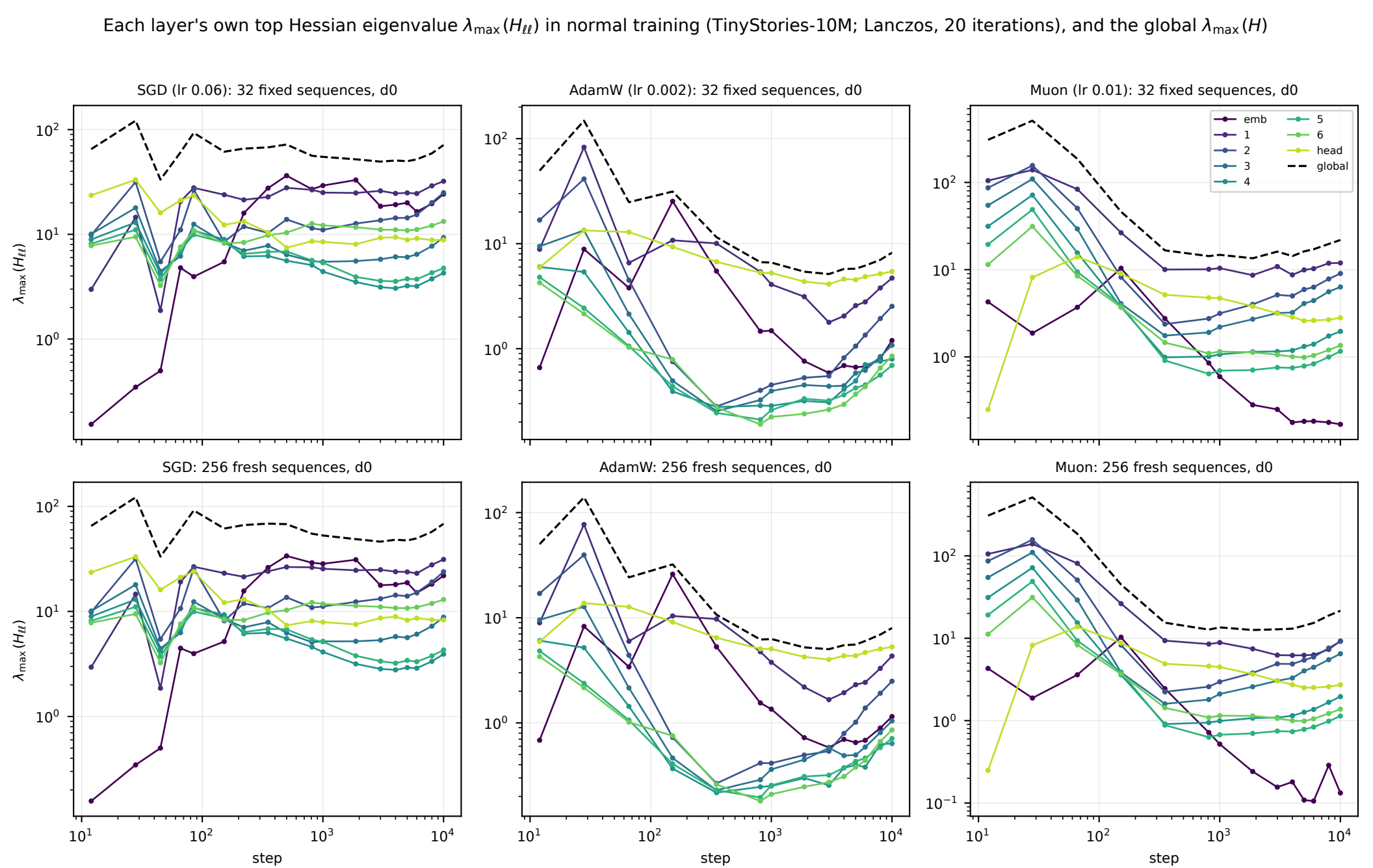
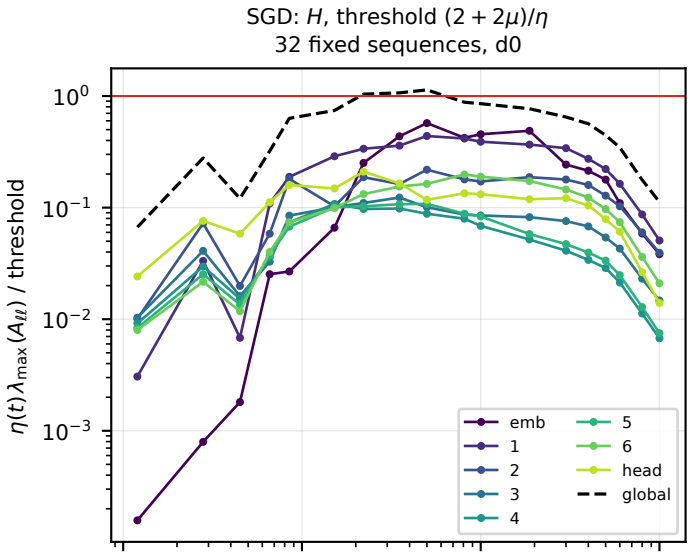
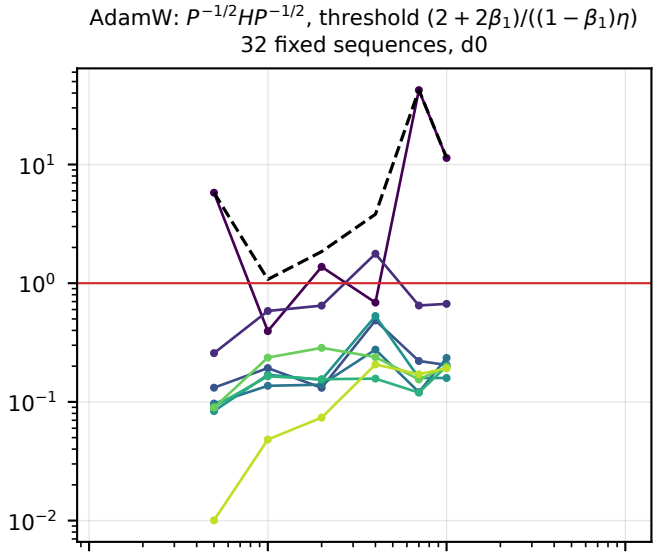


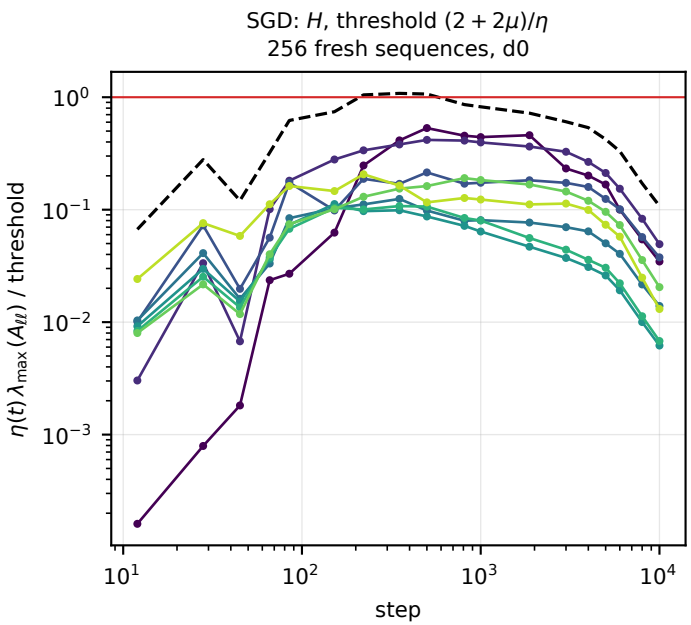
Geometric mean over the data orders at each checkpoint. The 32 fixed sequences (those of the saved sharpness) cover few vocabulary rows, which lowers the embedding block.

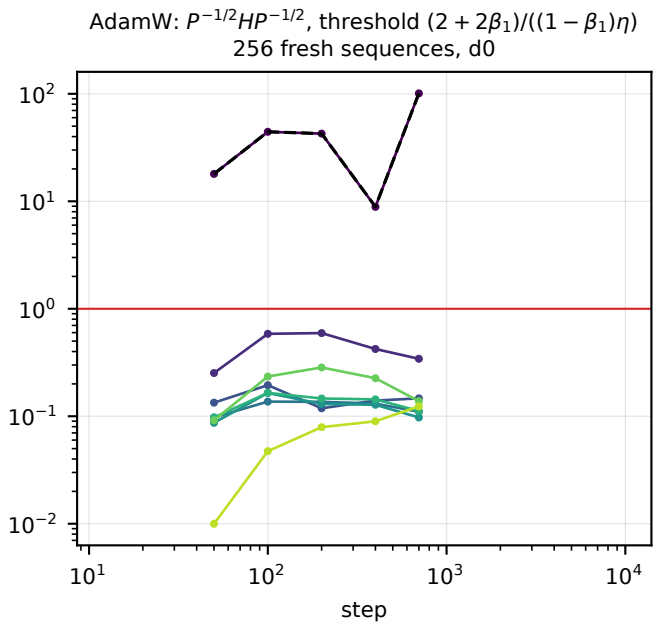
How close each layer alone is to the classic edge: $\eta(t)\,\lambda_{\max}(A_{\ell\ell})$ / threshold





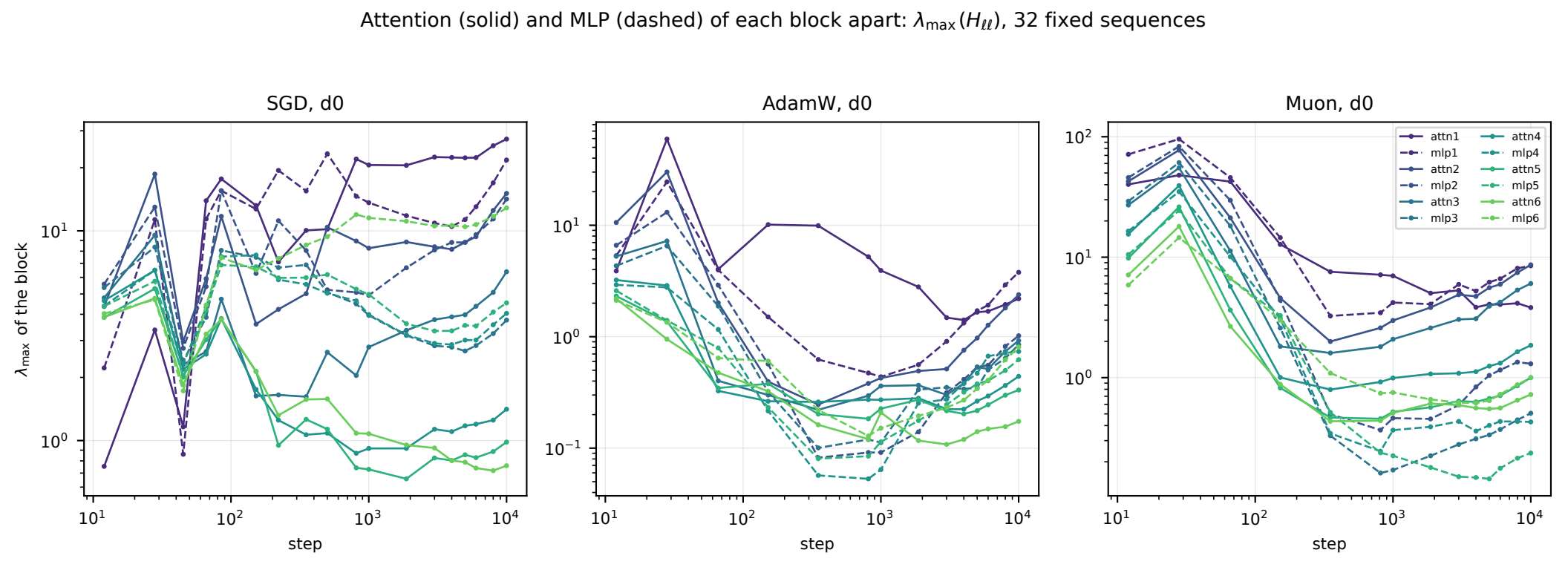
Muon: no classic threshold on H
(its hidden-layer step is normalized; its edge is
the spectral-norm sharpness). Raw values on page 1.





Muon: no classic threshold on H
(its hidden-layer step is normalized; its edge is
the spectral-norm sharpness). Raw values on page 1.

AdamW: P (from the optimizer state) only at the fork checkpoints of data order 0, steps 50-4000; no optimizer state was saved later or for the other orders. Red line: the classic edge.


Attention (solid) and MLP (dashed) of each block apart: $\lambda_{\max}(H_{\ell\ell})$, 32 fixed sequences
SGD, d0
AdamW, d0
Muon, d0
$\lambda_{\max}$ of the block
step
step
step
attn1
mlp1
attn2
mlp2
attn3
mlp3
attn4
mlp4
attn5
mlp5
attn6
mlp6

Summary: medians over the checkpoints of the geometric means over data orders

	layer	lam_ll <=351	lam_ll >=811	lam_ll/ global	fresh/ fixed	max eta*lam_ll/thr (32 fixed)	at step	max, fresh (256)
SGD	emb	4.37	20	0.38	0.95	0.57	500	0.53
	1	21	25	0.48	0.97	0.44	500	0.42
	2	10.7	14.3	0.28	0.98	0.22	500	0.21
	3	8.07	6.06	0.12	0.96	0.12	351	0.12
	4	7.84	3.5	0.06	0.96	0.11	152	0.11
	5	7.59	3.92	0.07	0.97	0.11	500	0.11
	6	8.29	11.7	0.22	0.99	0.20	811	0.19
	head	18.6	8.85	0.15	0.97	0.21	221	0.21
	all	65.5	52	1.00	0.97	1.13	500	1.08
AdamW	emb	5.48	0.761	0.12	0.98	42.33	700	100.95
	1	10.1	3.13	0.54	0.92	1.77	400	0.59
	2	4.5	0.819	0.14	0.97	0.49	400	0.19
	3	2.14	0.451	0.09	0.96	0.28	400	0.14
	4	1.43	0.411	0.07	0.87	0.53	400	0.16
	5	1.06	0.363	0.06	1.00	0.20	1000	0.16
	6	1.03	0.295	0.05	1.01	0.29	200	0.28
	head	9.32	4.85	0.79	0.97	0.21	400	0.12
	all	31.4	6.18	1.00	0.97	42.53	700	100.96
Muon	emb	3.7	0.183	0.01	0.88	-		-
	1	83.6	10.3	0.61	0.85	-		-
	2	50.5	5.14	0.35	0.96	-		-
	3	29.3	3.23	0.23	0.98	-		-
	4	15.6	1.18	0.08	0.96	-		-
	5	9.43	0.756	0.05	0.99	-		-
	6	8.48	1.1	0.07	1.00	-		-
	head	8.15	2.86	0.20	0.97	-		-
	all	186	16.1	1.00	0.93	-		-

lam_ll: lambda_max(H_ll) on the 32 fixed sequences, raw H for every optimizer, steps <= 351 and >= 811. lam_ll/global: its ratio to the global lambda_max(H) (<= 1 by interlacing).
fresh/fixed: lambda_max(H_ll) on 256 fresh sequences over that on the 32 fixed ones (d0). max eta*lam_ll/thr: SGD on H, AdamW on P^-1/2 H P^-1/2 (d0, steps 50-4000); 1 is the classic edge.